

Exercise 1: Electrical Circuits

Introduction

In this exercise, you will build and test five different electrical circuits, making observations and measurements along the way. The exercise is divided into two tasks, each beginning with a base circuit that you progressively modify through a series of subtasks.

Record your work as you go — take photos of each circuit configuration and note down all measurements. You will need this material when writing up your portfolio report.

All components and instruments can be obtained from the lab tutor. Each task lists the components you need for each circuit. In addition to those, you will need the *Electronic Box* toolbox throughout the entire exercise.

Important: If you are unsure about anything at any point, ask the tutor!

Few General Rules

- By default, **use the laboratory power supply from the assigned *Electronics Box***, unless specified otherwise in the manual.
- **Always switch off and disconnect power before modifying or rewiring a circuit.** Never change connections while the circuit is live.
- **If anything smells, sparks, or gets unexpectedly hot — immediately disconnect power and inform the tutor before doing anything else.**
- **Double-check your wiring against the schematic before applying power** for the first time. Pay particular attention to polarity — connecting a component backwards can damage it permanently.
- Ensure that **all your circuits have one common ground.**
- **Never exceed the specified supply voltages**, and be considerate with limiting current on the supply — if you have to increase the limit more and more, it's a red flag.
- If you are **unsure about anything** at any point, **ask the tutor before** powering your circuit.

Task 1 – LED Control Circuit

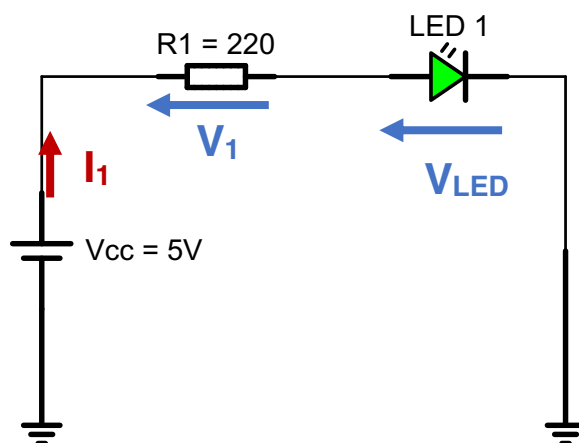
In this task, you will start by building a simple circuit to light up an LED. Through the subtasks, you will consecutively add more control over the circuit, introducing a switch and then a potentiometer to adjust the brightness.

Components required:

- Resistors: 100 Ω , 220 Ω , 1.0 k Ω , 4.7 k Ω
- Green LED
- 1 k Ω potentiometer
- 2-position switch
- Breadboard
- Set of male-male jumper wires

Task 1.1 – Simple LED Circuit

Compose the circuit as shown in the schematic:



Observe the behaviour of the LED. Use the multimeter to measure V_1 and I_1 .

In the following steps, replace R_1 with resistors of 1.0 k Ω , 4.7 k Ω and repeat the steps.

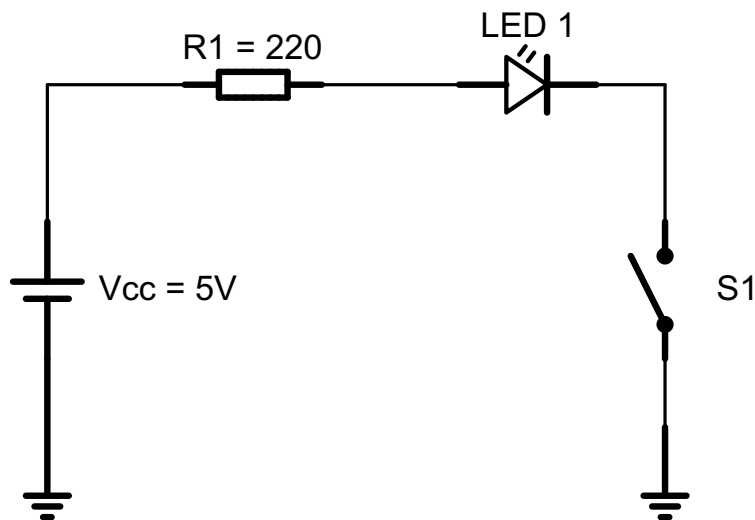
Complete the following table:

R_1 [Ω]	Measured V_1 [V]	Measured V_{LED} [V]
220		
1000		
4700		

What have you observed? What's the voltage drop on R_1 ? Does it change? Does changing the R_1 affect the LED?

Task 1.2 – Switchable LED Circuit

Compose the circuit as shown in the schematic:

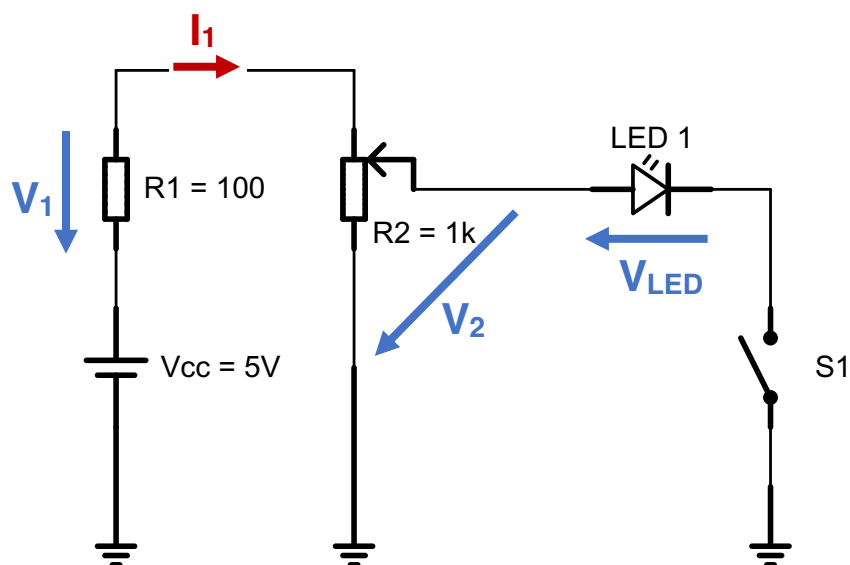


Use the switch. Observe the behaviour of the LED. Try connecting the switch in the opposite direction. What have you observed? Include your observations in your portfolio entry.

Task 1.3 – Dimmable LED Circuit

Compose the circuit as shown in the schematic:

Remember: middle pin of the potentiometer is the wiper: the variable output.



Operate the potentiometer and observe the behaviour of the LED.

Choose 4 positions of the potentiometer where you can observe: a) LED at full brightness; b) dimmed LED; c) LED is OFF.

If the regulation is difficult to achieve, you can exchange R1 with a 220 Ω resistor.

Use the multimeter to measure voltages.

Complete the following table:

Position	V_{LED} [V]	V_2 [V]
a) full brightness		
b) dimmed		
c) OFF		

What have you observed? How does the LED behave as you change the position of the potentiometer? How do the voltages change? What's the characteristics of this relationship?

Include the measurements and your observations in your portfolio entry.

Remember to photograph or record short videos of your circuits in operation.

Task 2 – Transistor Switch Circuit

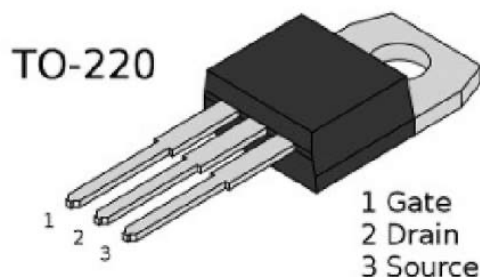
In this task, you will use a circuit with NPN MOSFET transistor to turn on/off and control the brightness of an LED strip using PWM (Pulse Width Modulation). Instead of a simple on/off switch, the gate is now driven by a PWM signal — rapidly switching the MOSFET on and off at a set frequency and duty cycle. The control side runs on 5V and the load side on 12V, with a common ground shared between them. Through the subtasks, you will explore how duty cycle affects perceived brightness, and how frequency determines whether the strip appears to flicker or glow steadily.

Components required:

- Resistors: 100 Ω , 10 k Ω
- PWM Signal Generator + cable
- Power-supply USB module
- NPN MOSFET IRLZ44N
- 2-position switch
- 12V LED Strip
- Breadboard
- Set of male-male jumper wires
- Few male-female jumper wires

In this part of the exercise, we will be using the transistor, with the following pinout:

NPN MOSFET IRLZ44N



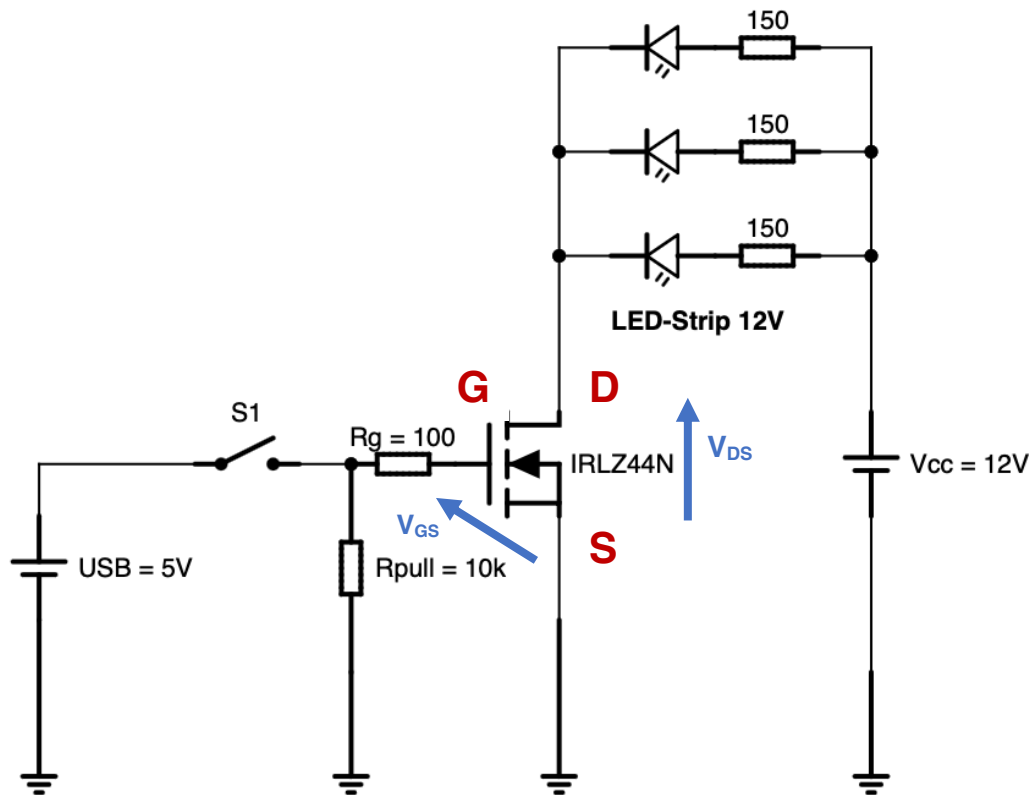
You will also use one of the two USB-driven power sources as your 5V DC source.

If you're using **SOLDERED Breadboard Power Supply**, use **VOUT2 output switched in 5V output mode**. Power the board using USB-C cable and power adapter.

If you're using **HW193 USB adapter**, plug it directly to the USB power adapter and transfer **5V and GND** to your breadboard using male-female jumper cables.

Task 2.1 – Switchable LED Strip

Compose the circuit as shown in the schematic:



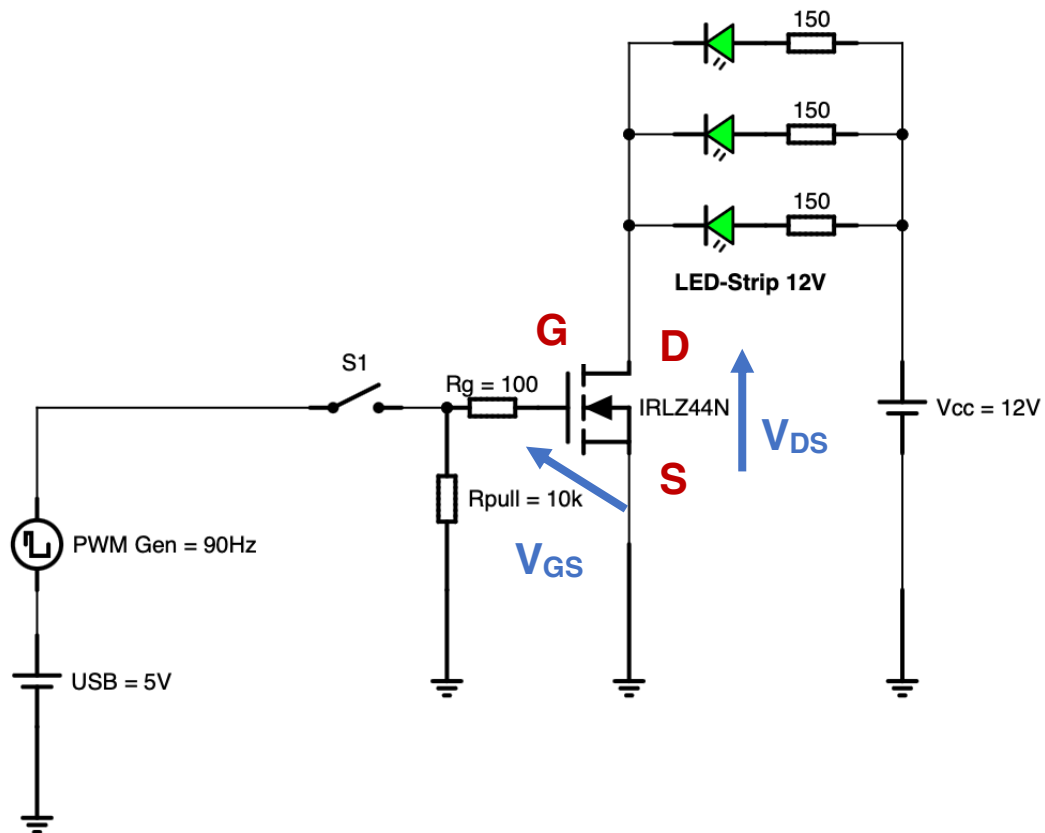
Do not touch the transistor while the circuit is powered and shortly after – the element may heat up!

Switch the circuit. What have you observed?

Which voltage are you controlling with the switch? In the portfolio, explain the principle of operation of the transistor switch. Include your observations in your portfolio entry.

Task 2.2 – Dimmable LED Strip

Compose the circuit as shown in the schematic:



The generator needs to be powered from your **USB power board**.

To do so:

- **power off the USB board** – pull the USB cable away/plug off the adapter
- connect **5V or VOUT2 pin** from **USB board** to **VIN+** pin on the **PWM generator**
- connect **GND pin** from **USB board** to **VIN-** pin on the **PWM generator**
- connect **PWM output** pin to **S1** in your circuit
- connect **GND pin on output** side to the **GND** of your circuit

Switch on the circuit and operate the settings of the PWM generator.

Set the frequency at **f = 90 Hz**.

A) First, you will adjust the Duty Cycle.

Observe the LED Strip for 5 different settings of the **Duty Cycle (D)**:

- a) D = 2%
- b) D = 15%
- c) D = 40%
- d) D = 75%
- e) D = 100%

What have you observed? How does the LED-Strip behave as you change the duty cycle setting? What is happening? What's the characteristics of this relationship?

Compare your observations with the dimmable LED circuit. What have you observed? Describe your observations in your portfolio entry.

B) Second, you will adjust the **Switching Frequency**.

Switch the circuit and operate the settings of the PWM generator.
Keep the duty cycle at **D = 0.5**.

Observe the LED Strip for 4 different settings of the **Frequency (f)**:

- a) $f = 5 \text{ Hz}$
- b) $f = 25 \text{ Hz}$
- c) $f = 45 \text{ Hz}$
- d) $f = 100 \text{ Hz}$

You can also experiment with other frequencies.

What have you observed? How does the LED-Strip behave as you change the frequency setting? What is happening, and when does it stop? Describe your observations in your portfolio entry.

Remember to photograph or record short videos of your circuits in operation.